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REVIEW

A Review of Community-Based Frailty and Malnutrition Screening in Older Adults: Current Interventions and Tools

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Abstract

Frailty and malnutrition are prevalent and interrelated conditions among community-dwelling older adults, contributing to functional decline, reduced quality of life, and increased healthcare utilization. Early identification through community-based screening offers an important opportunity for prevention; however, screening practices and linked interventions remain heterogeneous across settings. This structured narrative review synthesized peer-reviewed studies published between January 2021 and February 2026 identified through the Scopus database. Eligible studies focused on adults aged≥60 years and examined either community-based frailty or malnutrition screening tools or community interventions that incorporated screenings. Data were synthesized descriptively across two analytical components: screening tools and screening-informed interventions. A total of 42 studies were included. Screening tools ranged from low-resource anthropometric measures and brief questionnaires to structured nutritional indices and multidomain frameworks. Interventions commonly integrated screening with nutrition-focused strategies, physical activity, education, or multidomain lifestyle programs delivered in community settings. Recent studies increasingly emphasized feasibility, early risk identification, and linkage between screening and preventive action. Community-based frailty and malnutrition screening is increasingly used as a gateway to preventive interventions rather than as an isolated assessment. Aligning feasible screening tools with actionable intervention pathways is critical for advancing healthy aging strategies. Future research should prioritize long-term outcomes, implementation sustainability, and system-level integration within community health services.

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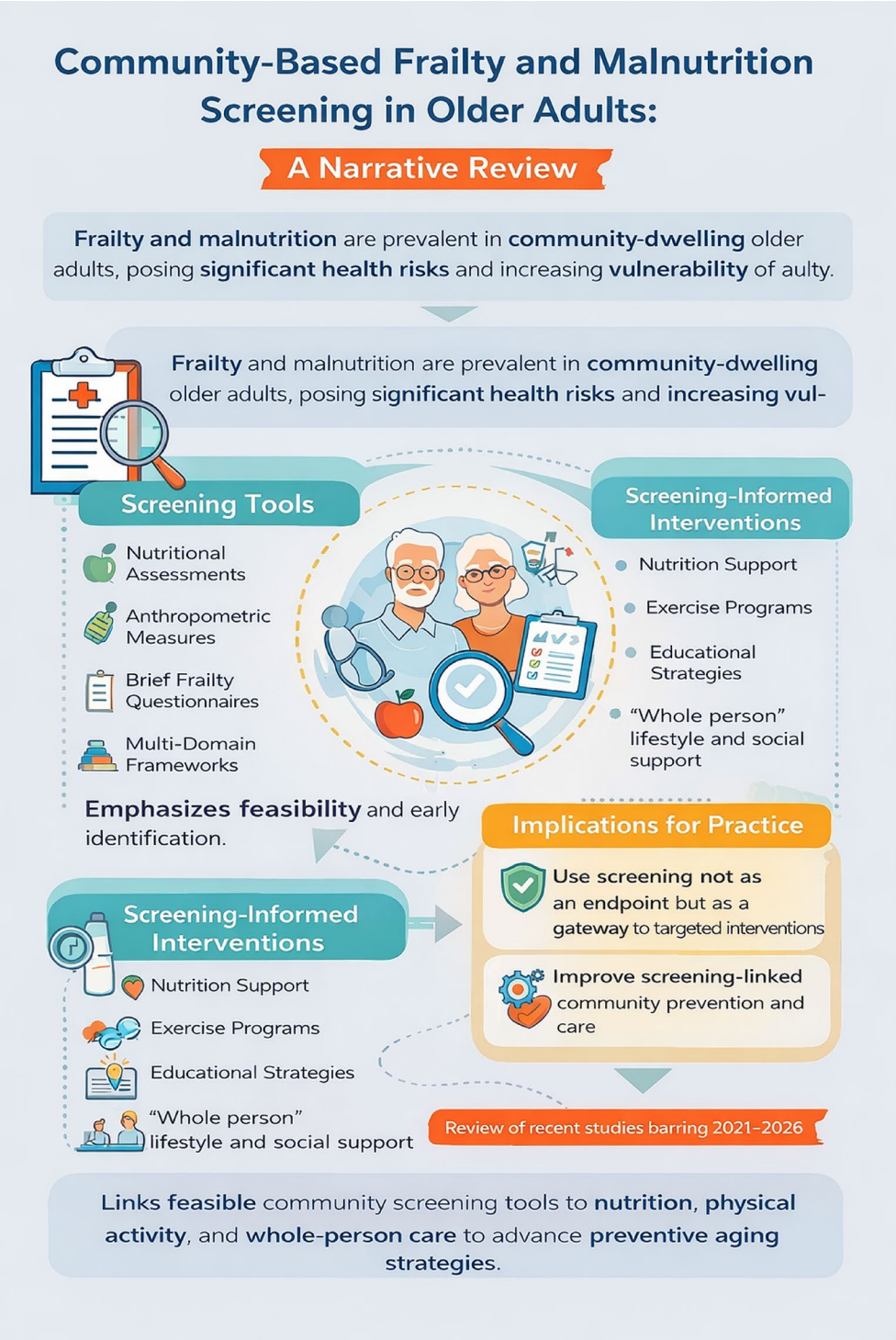
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Graphical Abstract



Keywords Frailty · Malnutrition · Older adults · Interventions · Tools

Introduction

Population aging presents a major public health challenge worldwide, with increasing numbers of older adults living with complex health needs in community settings. Among

these, frailty and malnutrition are particularly prevalent and often coexisting conditions that contribute to functional decline, disability, hospitalization, and mortality [1,2]. Importantly, both conditions are potentially modifiable, especially when identified early, making them priority targets for community-based prevention strategies.

Frailty is commonly conceptualized as a state of increased vulnerability resulting from age-related decline across multiple physiological systems, leading to reduced resilience to stressors [3,4]. Malnutrition, including undernutrition and inadequate protein or energy intake, is both a contributor to and consequence of frailty, particularly among community-dwelling older adults [5,6]. Despite their clinical significance, frailty and malnutrition frequently remain under-recognized in community and primary care settings, where time, resources, and specialist expertise may be limited.

In response, a wide range of community-based screening tools have been developed to facilitate early identification of frailty and nutritional risk. These tools vary considerably in scope, complexity, and feasibility, ranging from brief questionnaires and anthropometric measures to structured indices and multidomain frameworks [7]. Parallel to these developments, community-based interventions increasingly incorporate screening results to guide nutrition support, physical activity, education, and integrated care pathways [1,8]. However, evidence remains fragmented, and there is limited synthesis that focused on how screening tools are applied in community contexts and how they inform subsequent interventions.

Therefore, the objective of this review was to synthesize recent evidence on community-based frailty and malnutrition screening in older adults, with a dual focus on (i) screening tools used in community settings and (ii) community-based interventions that incorporate screening as part of prevention or service delivery. By examining both screening feasibility and actionability, this review aims to inform community health practice and support the development of integrated, prevention-oriented aging strategies.

Methods

Given the growing emphasis on early identification and prevention of frailty and malnutrition in community settings, a structured synthesis of recent evidence was undertaken to capture both screening practices and linked interventions among older adults. The methodological approach was designed to reflect real-world community health contexts, where feasibility, scalability, and actionability are central to screening and prevention strategies.

Study Design

This review employed a two-part structured narrative review design to synthesize heterogeneous evidence across community health domains [9]. Narrative and structured reviews are particularly suited to public health topics where study designs, settings, and outcome measures vary substantially and where implementation context is central to interpretation. The review focused on community-based frailty and malnutrition screening in older adults and was organized into two analytic components:

(i) identification and description of frailty and malnutrition screening tools used in community settings, and.
(ii) synthesis of current community-based interventions that incorporated frailty or malnutrition screening as part of service delivery or prevention strategies.

Data Source and Search Strategy

The literature search was conducted using Scopus, selected for its broad coverage of public health, gerontology, primary care, and community health research and its suitability for multidisciplinary reviews [10]. The dataset comprised 351 peer-reviewed records published between January 2021 and February 2026. Searches were restricted to English-language publications.

Search terms combined four conceptual domains:
(1) population (older adults),
(2) condition (frailty and/or malnutrition),
(3) screening or assessment, and.
(4) community setting.

Searches were applied to article titles, abstracts, and author keywords to maximize sensitivity. Records were exported in CSV format for systematic screening and data extraction.

Eligibility Criteria

Studies were included if they met all of the following criteria:

- Focused on adults aged 60 years or older;
- Conducted in community-based, non-institutional settings, including primary care, home care, senior centers, public health programs, or community outreach initiatives;
- Addressed frailty or malnutrition screening tools (Part 1) or interventions that incorporated screening (Part 2);
- Published as peer-reviewed journal articles.

Studies were excluded if they:

- Were conducted exclusively in acute hospital or inpatient settings, as these differ substantially from community contexts in screening feasibility and implementation.
- Focused solely on diagnostic biomarkers or laboratory measures without a screening or assessment component.
- Were editorials, conference abstracts, protocols, commentaries, or other non-peer-reviewed formats.

These criteria were selected to ensure relevance to community health practice and alignment with prevention-oriented aging research.

Study Selection

All records were imported into a reference management environment, and duplicate records were removed prior to screening. Titles and abstracts were screened for relevance to community-based frailty or malnutrition screening. Full texts were reviewed when eligibility could not be determined from abstracts alone. Screening decisions were guided by the predefined inclusion and exclusion criteria.

The study selection process followed PRISMA 2020 reporting principles, with records categorized into identification, screening, eligibility, and inclusion stages to enhance transparency and reproducibility [11].

Data Extraction and Synthesis

For Part 1, data were extracted on screening tools type, assessment domains (e.g., nutritional, physical, multidomain), and applicability within community settings. For Part 2, extracted data included intervention type, delivery setting, target population, and the extent to which screening results informed subsequent actions such as referral, education, or support.

Given heterogeneity in study designs, outcomes, and implementation contexts, findings were synthesized descriptively rather than quantitatively, consistent with best practices for narrative and public health reviews [12]. Screening tools and interventions were grouped thematically to facilitate comparison across community contexts and to identify patterns relevant to practice and policy. Figure 1. PRISMA 2020 flow diagram of the study selection process. Records were identified through Scopus database searching ($n=351$). Following title and abstract screening, 289 records were excluded. Full-text articles were assessed for eligibility ($n=62$), of which 20 were excluded for non-community-based settings, lack of a screening component, or publication type. A total of 42 studies were included in the qualitative synthesis.

Results

Overview of Included Studies

The included literature spanned January 2021 through February 2026 and addressed either (1) community-based frailty and/or malnutrition screening tools (Part 1) or (2) community-based interventions incorporating screening (Part 2), with some studies contributing evidence relevant to both components. Consistent with the evolving emphasis on community prevention and healthy aging, the included studies increased in frequency over time, with many published between 2023 and 2025. Studies were conducted across multiple regions, including East Asia (e.g., Korea, China, Singapore, Japan), Europe, and North America, underscoring broad international interest in scalable screening and intervention strategies for aging populations.

Part 1: Frailty and Malnutrition Screening Tools

The included studies demonstrated substantial diversity in community-feasible screening approaches, ranging from brief anthropometric measures and questionnaires to composite indices and multidomain functional screening tools (Table 1). Several studies evaluated nutritional screening measures as practical entry points for identifying frailty-related vulnerability. For example, Desiana (2025) [13] examined the combined use of Mini Nutritional Assessment (MNA) scores and body mass index, alongside a community frailty checklist, to identify frailty risk among community-dwelling older adults, supporting the feasibility of nutrition-based screening for early frailty detection. Similarly, Kim (2024) [14] evaluated mid-upper arm circumference (MUAC) within a large community cohort and demonstrated its utility as a pragmatic, low-resource anthropometric marker for identifying physical frailty in community-dwelling older adults.

A second cluster of studies focused on structured indices and multidomain screening tools to enhance risk stratification in community populations. Previous findings had reported on the ENIGMA nutritional risk index, providing evidence of its validity and associations with adverse health outcomes in community-dwelling older adults [15,16]. These findings highlight growing interest in index-based screening approaches that may be scalable beyond traditional nutritional tools. Beyond nutritional risk, community screening increasingly extended to broader functional vulnerability. Lu (2023) [17] evaluated the diagnostic performance of the World Health Organization Integrated Care for Older People (WHO ICOPE) screening tool for identifying intrinsic capacity decline, emphasizing the feasibility of

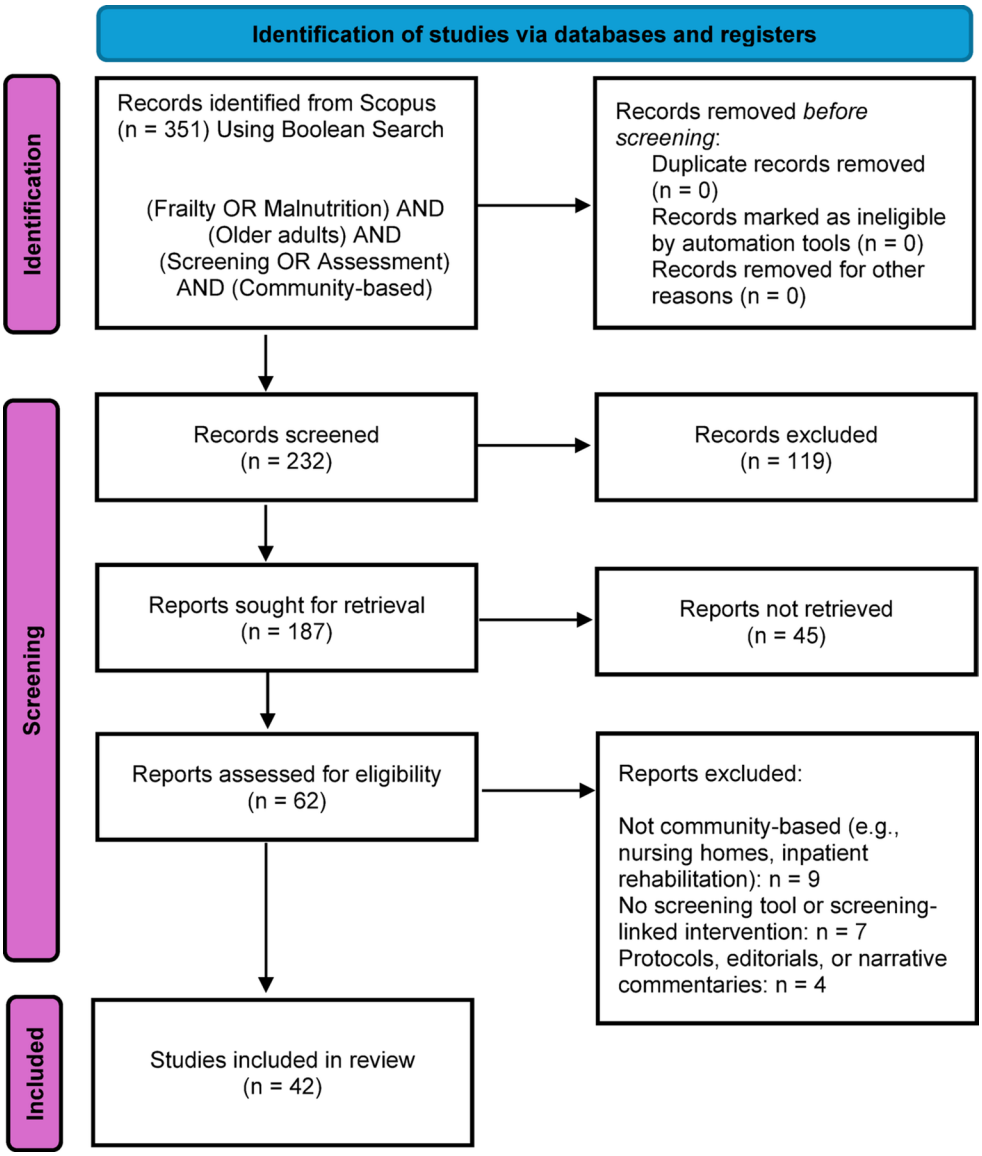


Fig. 1 The PRISMA flow diagram detailing the screening and selection process of literature

brief multidomain screening that is relevant to frailty prevention in community care settings.

Frailty-specific case-finding tools were also well represented. Okyar-Bas (2023) [18] validated a Turkish version of PRISMA-7, supporting its use as a brief case-finding instrument for identifying older adults living with frailty in community or outpatient contexts [19]. Complementing these single-tool studies, one study had compared multiple frailty, malnutrition, metabolic, and inflammatory indices within a longitudinal community cohort, illustrating increasing use of combined risk profiling to predict multimorbidity progression and mortality among community-dwelling older adults [20]. Finally, Edwards (2022) [21]evaluated the PaperWeight Armband, a MUAC-based screening device, demonstrating the feasibility of extremely low-resource undernutrition case-finding paired with brief follow-up action in community outreach settings. Collectively, these studies indicate a shift toward a spectrum of screening tools, which ranging from simple anthropometry to multidomain frameworks, that selected primarily for feasibility and scalability in non-specialist community settings (Table 1). This trend is further supported that biopsychosocial frailty tools suitable for community populations and advocated for broader conceptualization of frailty beyond purely physical criteria [7].

Part 2: Community-Based Interventions

A substantial subset of included studies described community-based interventions that explicitly incorporated frailty and/or malnutrition screening into eligibility determination, intervention tailoring, or outcome evaluation (Table 2). Nutrition-focused interventions were among the most commonly reported and frequently used screening to

Table 1 Community-Based Frailty and Malnutrition Screening Tools Among Older Adults

Study	Study type	Community setting	Population	Screening tool / measure	What the tool measures	Key results relevant to community screening (as reported)
Desiana (2025) [13]	Cross-sectional/ community study	Community (Indonesia)	Community-dwelling older adults	MNA score + BMI (with Kihon's Frailty Checklist)	Nutritional status + frailty status	Evaluated the predictive/associative value of MNA and BMI for frailty status using a community frailty checklist; supports feasibility of nutrition screening to flag frailty risk in the community
Kim (2024) [14]	Cohort-based analysis (KFACS)	Community-dwelling cohort	Community-dwelling older adults (Korea)	Mid-upper arm circumference (MUAC)	Anthropometric proxy for muscle/nutritional reserves; frailty risk indicator	Tested MUAC as a screening tool for identifying physical frailty in community-dwelling older adults; supports MUAC as a pragmatic, low-resource community screen
Wei (2024) [15]	Observational study	Community (China)	Chinese community-dwelling older adults	ENIGMA (Elderly Nutritional Indicators for Geriatric Malnutrition Assessment)	Nutritional risk index	Examined nutritional risk using ENIGMA and links to adverse outcomes; supports use of structured nutritional indices for risk stratification in community settings
Ng (2022) [16]	Validation/ construct study	Community-based cohort	Community-dwelling older persons	ENIGMA index (external evaluation)	Nutritional risk/ geriatric malnutrition risk	Reported concurrent, construct, and predictive validity of ENIGMA in an external community cohort; supports index-based nutritional screening in community populations
Lu (2023) [17]	Diagnostic performance study	Community/ primary care-type screening context	Older adults screened for intrinsic capacity decline	WHO ICOPE screening tool	Intrinsic capacity decline domains (multi-domain function)	Reported diagnostic performance of the ICOPE screening tool to identify decline; supports feasibility of brief multi-domain screening relevant to frailty prevention in community care
Okyar-Bas (2023) [18]	Validation study	Community / outpatient context	Older adults (Turkey)	PRISMA-7 (Turkish version)	Case-finding for frailty	Reported validation of PRISMA-7 as a case-finding instrument for detecting older adults living with frailty; supports brief frailty case-finding in community settings
Lee (2021) [19]	Observational association study	Community-dwelling (Korea)	Korean older adults	FRAIL scale	Frailty risk (brief questionnaire)	Reported associations between FRAIL scale and geriatric syndromes/health outcomes; supports FRAIL as a feasible community screen linked with broader vulnerability
Cheong (2023) [20]	Longitudinal cohort analysis	Community cohort (Singapore)	Older adults in a longitudinal aging study	Frailty/malnutrition/metabolic/ inflammatory indexes	Multi-index risk profiling	Compared multiple frailty and malnutrition-related indexes with multimorbidity incidence/progression and mortality impact; supports index-based stratification approaches in community aging cohorts
Edwards (2022) [21]	Community evaluation / brief intervention study	Community outreach settings	Older people at risk of undernutrition	PaperWeight Armband (MUAC-based)	Undernutrition risk case-finding	Evaluated feasibility of using a very low-resource MUAC-based tool to identify undernutrition risk in the community, paired with brief follow-up action
De Luca (2022) [7]	Narrative review	Community-dwelling focus	Community-dwelling older adults	Biopsychosocial frailty tools (multiple)	Physical, psychological, and social frailty dimensions	Summarized multidimensional frailty tools used in community populations, supporting broader conceptualization of frailty beyond physical phenotype in community screening

identify individuals at risk prior to intervention delivery. For example, latest findings reported outcomes of a community nutrition-focused program in which frailty screening was used to assess and monitor changes in frailty-related status, illustrating a nutrition-led approach to community frailty prevention [22]. Similarly, Struszczak (2025) [23] evaluated a community intervention involving a daily high-protein, high-energy meal targeted to malnourished or at-risk older adults, with nutrition status assessment directly linked to intervention delivery and evaluation. Previously, study had assessed the impact of a nutritionally complete oral nutritional supplement in a community context, further reflecting integration of screening with targeted nutritional intervention [24].

Interventions integrating nutrition and physical activity were increasingly represented in recent literature. Zhou (2025) [25] examined a theory-driven exercise intervention among pre-frail and frail community-dwelling older adults

Table 2 Community-Based Interventions Incorporating Frailty and Malnutrition Screening in Older Adults

Study	Study type	Community setting	Population	Intervention type	Screening integration (frailty/malnutrition)	Key results relevant to community practice (as reported)
Cano-Gutierrez (2025) [22]	Intervention evaluation	Community	Older adults	Nutrition-focused program	Frailty screening used to assess/change frailty status	Reported program outcomes on frailty-related status; illustrates nutrition-led community approach
Zhou (2025) [36]	Intervention study	Community	(Pre)frail community-dwelling older adults	Theory-driven exercise intervention	Baseline nutritional status used to evaluate differential effectiveness	Examined how nutritional status modifies exercise intervention response in (pre)frail older adults
Buhl (2025) [37]	Randomised controlled trial	Community-dwelling	Adults ≥ 80 years	Whole-food dairy protein supplementation	Screening/eligibility includes community-dwelling ≥ 80; nutrition-related targeting	Reported success/acceptability outcomes for personalised protein supplementation
Pöyhönen (2025) [28]	Post hoc analysis of RCT	Community-based trial context	Older adults	Multidomain lifestyle intervention	Frailty outcome used to assess short/long-term effect	Examined short- and long-term frailty effects of a multidomain lifestyle intervention
Ng (2025) [29]	Community intervention study	Community	Pre-frail older adults	Community nutrition education	Pre-frail screening; sarcopenia markers assessed	Evaluated nutrition education effects on intake and sarcopenia-related markers in pre-frail adults
Struszczak (2025) [23]	Intervention study	Community-dwelling	Malnourished/at-risk older adults	Daily high-protein/high-energy meal	Malnutrition status assessed; nutrition screening linked to intervention	Reported effects on physical and psychological wellbeing in malnourished community-dwelling older adults
Sen (2025) [32]	Systematic review	Community/public health	Older adults	Interventions targeting social isolation + food insecurity	Food insecurity/malnutrition risk identification	Synthesized interventions targeting social isolation and food insecurity relevant to nutrition vulnerability in older adults
Ma (2024) [31]	Feasibility / implementation study	Community	Older adults	WHO ICOPE implementation feasibility	ICOPE screening embedded into program	Evaluated feasibility of implementing ICOPE approach in community settings
Pedrinolla (2023) [38]	Randomised controlled trial	Community	Older adults	Vitamin D + other components (trial-based)	Frailty/sarcopenia-related screening criteria used	Reported results of RCT combining vitamin D with other strategies relevant to frailty prevention
Ward (2023) [27]	Trial (BMJ Open)	Community	Older adults	Protein-enriched Mediterranean diet approach	Nutrition screening/risk stratification informs intervention	Examined effects of protein-enriched Mediterranean diet strategy on relevant aging outcomes
Casals (2023) [30]	Educational intervention study	Community	Older adults	Educational intervention	Frailty status measured pre/post	Reported effects of education on frailty-related outcomes in older adults
Tufan (2023) [39]	Observational/clinical-community study	Community-dwelling	Older adults	(Intervention-related comparison)	Frailty stratification used	Evaluated how frailty status influences relevant outcomes among community-dwelling older adults
Millan-Domingo (2022) [40]	Real-life program outcomes	Community	Older adults	Multicomponent exercise intervention	Frailty screening/stratification used	Reported "real-life" outcomes from multicomponent exercise program
Tay (2022) [41]	Intervention analysis	Community	Older adults	Intrinsic capacity-focused approach	Intrinsic capacity screening considered	Reported that intrinsic capacity patterns were more explanatory than "intervention exposure" in outcomes

Table 2 (continued)

Study	Study type	Community setting	Population	Intervention type	Screening integration (frailty/malnutrition)	Key results relevant to community practice (as reported)
Yeung (2022) [24]	Intervention study	Community	Older adults	Nutritionally complete oral nutritional supplement	Malnutrition risk assessed and linked to ONS	Evaluated effect of nutritionally complete ONS in community context
Sunguya (2022) [42]	Intervention modeling/review	Population/community	Older adults	Targeted + population-wide interventions	Risk identification implied for targeting	Discussed targeted vs. population-wide intervention strategies relevant to undernutrition
Khor (2022) [43]	Systematic review	Community	Older adults	Nutrition interventions	Nutrition risk/assessment central	Synthesized effectiveness of nutrition interventions for community older adults
Edwards (2022) [21]	Community brief intervention evaluation	Community outreach	Older adults	PaperWeight Armband + brief action	MUAC-based screening triggers brief intervention	Evaluated screening-triggered brief intervention to identify undernutrition risk in community
Kim (2021) [44]	Intervention study	Community	Older adults	Social network intervention	Frailty/functional measures tracked	Reported reinforcement effects of social network intervention relevant to frailty prevention
Jyväkorpi (2021) [45]	Article	Community/older populations	Older people	Sarcopenia/physical frailty-related approach	Frailty/sarcopenia assessment used	Addressed sarcopenia and physical frailty in older populations relevant to community strategies
Matsuo (2021) [34]	Intervention study	Community/outpatient oral health context	Older adults	Oral hypofunction improvement program	Oral frailty screening/assessment	Reported improvements in oral hypofunction via comprehensive approach relevant to nutrition/frailty
Uemura (2021) [35]	Intervention study	Community	Older adults	Active learning program	Frailty-related outcomes assessed	Reported effectiveness of active learning program relevant to functional aging outcomes
Chaudhuri (2025) [33]	Needs assessment (program participation)	Community (Texas)	Older adults	Congregate meal participation factors	Meal program participation linked to nutrition vulnerability	Identified barriers/facilitators to meal program participation relevant to nutrition intervention reach

and demonstrated that baseline nutritional status modified intervention response, highlighting nutrition as a key determinant of functional benefit. Several trials focused on targeted dietary strategies to support function and frailty prevention, including whole-food dairy protein supplementation among adults aged 80 years and older (Buhl, 2025) and protein-enriched Mediterranean dietary approaches in which nutrition screening informed intervention stratification [26,27] Broader multidomain approaches were also evident; Pöyhönen (2025) [28] evaluated short- and long-term effects of a multidomain lifestyle intervention, using frailty outcomes as assessment endpoints, supporting the relevance of comprehensive intervention strategies in community-based prevention.

Multiple studies emphasized education, behavior change, and implementation feasibility. Previous findings had evaluated community-based nutrition education among pre-frail older adults, using pre-frailty screening and sarcopenia markers for assessment, reflecting prevention-oriented targeting before advanced decline [29,30]. Their reported outcomes of an educational intervention in which frailty status was measured before and after intervention, further supporting education as a feasible component of community frailty prevention strategies. In parallel, Ma (2024) [31] evaluated the feasibility of implementing the WHO ICOPE approach in community settings, demonstrating increasing prioritization of integrated models for intrinsic capacity and frailty prevention in real-world contexts.

Other interventions extended beyond conventional lifestyle programs to address social determinants and service uptake. Current findings also synthesized interventions targeting social isolation and food insecurity, highlighting growing recognition of upstream determinants influencing nutrition vulnerability and frailty risk among community-dwelling older adults [32]. Additionally, another study also identified barriers and facilitators to congregate meal participation among older adults in Texas, directly informing the reach and effectiveness of community nutrition programs

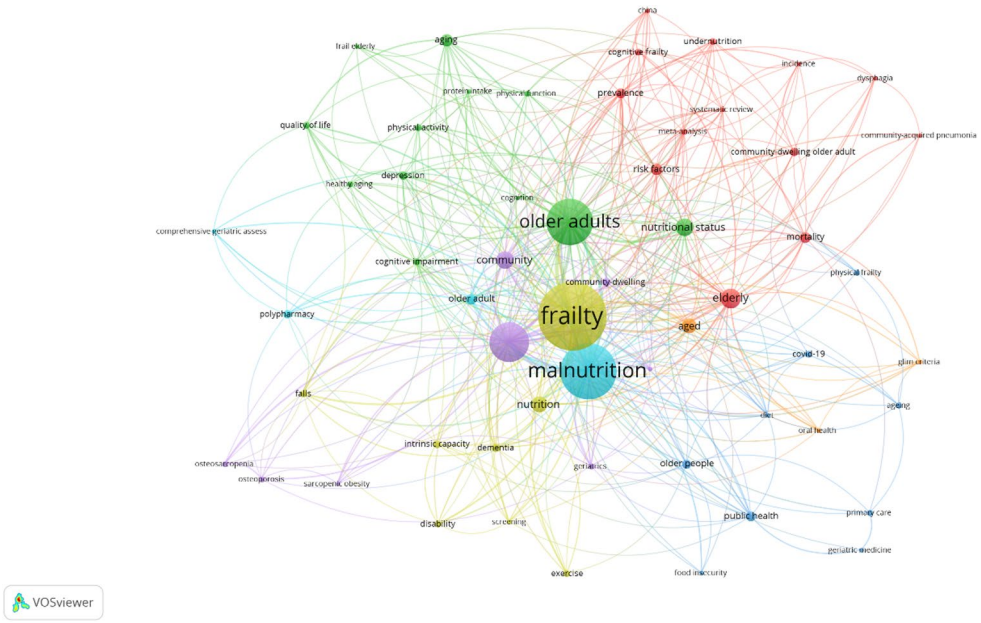


Fig. 2 Author keyword co-occurrence network of community-based frailty and malnutrition research. Node size represents keyword frequency and link thickness indicates co-occurrence strength. Colors denote thematic clusters across screening domains, interventions, outcomes, and community health contexts

[33]. Additional studies addressed alternative domains linked to frailty and nutrition, including oral hypofunction improvement programs where oral frailty screening was central [34] and active learning programs evaluating frailty-related outcomes [35]. Across these interventions, screening was consistently used not only for participant selection but also to guide referral pathways, tailor intervention intensity, and monitor intervention impact (Table 2).

Overall, the findings indicate that community-based frailty and malnutrition screenings are increasingly operationalized as a gateway to preventive action, rather than as an isolated assessment activity. Screening tools ranged from low-resource anthropometric measures to structured indices and multidomain functional frameworks, reflecting a continuum from feasibility-driven case-finding to comprehensive risk profiling. Interventions commonly linked screening to actionable strategies such as nutritional supplementation, dietary modification, structured physical activity, and multidomain lifestyle programs, with emerging attention to implementation feasibility and social determinants. Despite these advances, substantial heterogeneity persisted across tools, intervention models, and outcome measures, and relatively few studies addressed long-term sustainability or system-level impact, highlighting important gaps for future community-based frailty and malnutrition research.

Keyword Co-Occurrence Network and Thematic Clustering

Author keyword co-occurrence analysis identified frailty and malnutrition as the most central and highly interconnected concepts within community-based aging research, closely linked with *older adults* and *elderly* (Fig. 2). Surrounding these core terms, distinct thematic clusters reflected screening domains, intervention strategies, implementation contexts, and health outcomes.

A prominent cluster centered on nutritional status and undernutrition, incorporating terms such as *diet*, *protein intake*, and *GLIM criteria*, highlighting nutrition as a primary focus for community-based screening and prevention. Closely related, a physical frailty and sarcopenia cluster emphasized musculoskeletal vulnerability, including *physical frailty*, *sarcopenia*, *osteosarcopenia*, and *falls*, underscoring the importance of functional decline in community-dwelling older adults.

Contextual terms formed a clear community and public health cluster, with frequent co-occurrence of *community*, *community-dwelling*, *primary care*, and *public health*, reflecting the non-institutional settings emphasized in the literature. In parallel, a multidomain aging cluster incorporating *intrinsic capacity*, *cognition*, and *comprehensive geriatric assessment* indicated growing attention to holistic and functional screening frameworks.

Intervention-related keywords clustered around screening, prevention, and intervention, demonstrating that screening is commonly linked to actionable pathways rather than isolated assessment. Additional clusters captured lifestyle and physical activity, psychosocial and quality of life factors, and adverse outcomes such as *disability* and *mortality*, reflecting the multidimensional scope of community frailty and malnutrition research. Smaller clusters related to emerging and context-specific topics, including *COVID-19*, *food insecurity*, and *oral health*, highlighted evolving research priorities.

Discussions

This review synthesizes recent evidence on community-based frailty and malnutrition screening and related interventions among older adults, highlighting evolving practices that emphasize feasibility, prevention, and actionability within real-world community health contexts. Collectively, the findings demonstrate a shift away from isolated assessment toward screening-informed intervention pathways, reflecting broader preventive aging and population health frameworks.

Across included studies, a central theme was the prioritization of screening tools that are feasible, low-burden, and suitable for non-specialist delivery in community settings. Nutritional screening instruments such as the Mini Nutritional Assessment and anthropometric measures (e.g., MUAC) were frequently used as pragmatic entry points for identifying vulnerability, consistent with earlier evidence linking malnutrition and frailty risk in community-dwelling older adults [13,14]. Importantly, these tools require minimal equipment and training, supporting scalability in primary care, outreach, and home-based services.

At the same time, several studies demonstrated increasing interest in structured indices and multidomain screening frameworks, including ENIGMA and the WHO ICOPE tool [16,17]. These approaches extend beyond single-domain screening by capturing broader aspects of intrinsic capacity and cumulative risk. Such findings align with contemporary aging models that conceptualize frailty as a multidimensional process influenced by physical, nutritional, cognitive, and social factors [7]. However, while multidomain tools may offer improved risk stratification, their implementation may require additional training and system-level support, underscoring ongoing trade-offs between comprehensiveness and feasibility in community practice.

Brief frailty case-finding tools, such as PRISMA-7 and the FRAIL scale, were commonly reported and validated across diverse community populations [18,46]. Their consistent association with adverse outcomes reinforces their value as first-line screening instruments in settings where time and resources are limited. Notably, innovative low-resource approaches such as the PaperWeight Armband further illustrate how undernutrition risk can be identified even in outreach or volunteer-led contexts [21]. Together, these findings suggest that graduated screening pathways, beginning with simple case-finding and progressing to more detailed assessment when indicated, may be particularly suitable for community-based systems.

Beyond tool selection, the findings clearly indicate that screening is increasingly operationalized as a gateway to preventive intervention, rather than as an endpoint in itself. Most intervention studies explicitly integrated frailty or malnutrition screening into participant selection, tailoring, or outcome evaluation. Nutrition-focused interventions that including oral nutritional supplementation, high-protein meals, and dietary counselling were commonly targeted to individuals identified as at risk through screening [22,23,47]. This screening-to-action linkage is consistent with recommendations that early nutritional intervention can mitigate frailty progression and functional decline.

Moreover, interventions combining nutrition and physical activity were increasingly prominent, reflecting growing recognition of their synergistic effects. Studies demonstrated that baseline nutritional status can influence responsiveness to exercise interventions, reinforcing the importance of integrated screening to optimize intervention effectiveness [26,36]. Multidomain lifestyle interventions further supported this integrated approach, with evidence suggesting both short- and longer-term benefits for frailty-related outcomes [28].

In addition to clinical and lifestyle interventions, several studies emphasized education, implementation feasibility, and social determinants of health. Community nutrition education programs targeting pre-frail older adults demonstrated improvements in dietary intake and sarcopenia-related markers, highlighting the role of education as an early preventive strategy [29]. Implementation studies of the WHO ICOPE framework further underscored the growing emphasis on integrated, system-level approaches to intrinsic capacity and frailty prevention in community settings [31].

Importantly, the review also identified interventions addressing social isolation and food insecurity, factors increasingly recognized as upstream contributors to malnutrition and frailty risk [32]. Studies examining congregate meal participation and social network–based interventions illustrated how social engagement can reinforce nutritional and functional interventions, thereby enhancing sustainability and reach [33,48]. These findings highlight the necessity of embedding screening and intervention strategies within broader community and social care infrastructures.

Taken together, the evidence suggests that effective community-based frailty and malnutrition strategies require alignment between screening tools, intervention capacity, and implementation context. Simple, low-cost screening instruments can facilitate early identification, while more comprehensive tools may support targeted intervention planning when resources permit. Importantly, screening programs are most impactful when linked to actionable pathways, including referral, education, and multidisciplinary support.

However, despite promising trends, several gaps remain. Heterogeneity in screening tools, intervention designs, and outcome measures limits direct comparison across studies. Additionally, relatively few studies reported long-term

outcomes or system-level impact, such as sustained functional improvement, healthcare utilization, or cost-effectiveness. Future research should prioritize longitudinal evaluation and implementation science approaches to better understand how screening-informed interventions can be scaled and sustained within community health systems.

This review provides a focused synthesis of recent community-based evidence by restricting inclusion to peer-reviewed studies published between 2021 and 2026 and by emphasizing real-world community settings. Nevertheless, limitations should be acknowledged. The review was limited to English-language studies indexed in Scopus, which may have excluded relevant evidence from other databases or regions. Additionally, the narrative synthesis approach, while appropriate given heterogeneity, precludes quantitative comparison of effectiveness across interventions.

In conclusion, recent literature demonstrates a clear shift toward integrated, screening-informed approaches to frailty and malnutrition in community-dwelling older adults. Screening tools are increasingly selected for feasibility and scalability, while interventions emphasize nutrition, physical activity, education, and social support. Strengthening the linkage between screening and actionable community-based interventions, alongside attention to sustainability and equity, will be critical for advancing preventive aging strategies in population health contexts.

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Declarations

Competing interests None to disclose.

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